
The Job Sequencing and Tool Switching Problem

Part V: JGP + GSP Heuristic — Gap Analysis and Bounds

Scope and status. This note studies the additive gap $\text{gap}(I) = H(I) - Z_{\text{SSP}}^*(I)$ and the ratio $H(I)/Z_{\text{SSP}}^*(I)$ between the cost H of the two-phase *JGP + GSP* heuristic and the optimal uniform-SSP cost Z_{SSP}^* . What is **proved** here: (i) two lower bounds on Z_{SSP}^* — the grouping bound $Z_{\text{SSP}}^* \geq K^* - 1$ and the tool-counting bound $Z_{\text{SSP}}^* \geq |\bigcup_j T_j| - b$ (theorems 2.1 and 2.2), together with the classical $Z_{\text{SSP}}^* \leq b(K^* - 1)$; (ii) a computationally certified counterexample (the 6-job ring) with gap = 1 and ratio 4/3 (example 3.1); (iii) a construction with gap = g for every g , so the gap is *unbounded* (theorem 4.1); (iv) a directly computable gap bound $\text{gap} \leq H - \max(K^* - 1, |\bigcup_j T_j| - b)$ (proposition 5.1) and two sufficient conditions for zero gap (theorems 6.3 and 6.4); (v) a *grouping-exactness* theorem (theorem 6.1): Z_{SSP}^* equals the minimum GTSP cost over *all* feasible groupings, so the gap is exactly the price of using the fewest groups, and a sub-optimal grouping can strictly beat the heuristic (theorem 6.6). What is **conjectural**: the tight ratio bound $H/Z_{\text{SSP}}^* \leq 4/3$ for $b = 3$ (conjecture 5.4); it is *not* proved, and the previously attempted route via $\sum_k R_k \leq H/4$ is false and is retracted (section 5.2). All numerical assertions are verified by the scripts in `plans-genai/_verification/` (see `VERIFIED_FACTS.md`); reproduce with `python3 run_study.py`.

Contents

The JGP + GSP Heuristic

We work throughout with the *uniform* SSP: a single magazine of capacity b , jobs $j \in J$ with required tool sets $T_j \subseteq T$, $|T_j| \leq b$, and switch cost equal to the number of tool insertions along the processed sequence. We adopt the *free-initial-load* convention: the first $\leq b$ tools are loaded for free (equivalently, the sequence starts from a free dummy depot), so Z_{SSP}^* counts only the insertions made between consecutive jobs. For a fixed sequence the optimal loading is computed by KTNS [Tang and Denardo(1988)].

The Job Grouping Problem (JGP)

Definition 1.1 (Feasible group). A set of jobs $G \subseteq J$ is *feasible* if all its tools fit simultaneously: $|\bigcup_{j \in G} T_j| \leq b$.

Definition 1.2 (JGP as set covering/partitioning [Crama and Oerlemans(1994), Catanzaro et al.(2015)Catanzaro]). Let $\mathcal{S}$ be the family of feasible groups and $y_G \in \{0, 1\}$. The JGP is

$$Z_{\text{JGP}}^* = \min \left\{ \sum_{G \in \mathcal{S}} y_G : \sum_{G \ni j} y_G \geq 1 \quad \forall j \in J, y_G \in \{0, 1\} \right\},$$

the minimum number of feasible groups (“batches”) that cover J . We write $K^* = Z_{\text{JGP}}^*$.

Remark 1.3 (Hardness and exact solution). The JGP is NP-hard [Crama and Oerlemans(1994)]. It is solved exactly by the ARF MILP of Catanzaro et al. [Catanzaro et al.(2015)Catanzaro, Gouveia, and Labbé] and by Branch-and-Price [Otiai(2024)]; the pricing subproblem generates maximal feasible groups (a set-union knapsack).

Feasibility is a group property, not a pairwise one. A group is feasible iff the union of *all* its tool sets fits in b . This is stronger than pairwise compatibility: the three jobs $\{t_1, t_2\}, \{t_1, t_3\}, \{t_1, t_4\}$ ($b = 3$) are pairwise compatible (each pairwise union has size 3) yet jointly infeasible ($|\{t_1, t_2, t_3, t_4\}| = 4 > b$). Consequently K^* is *not* in general the chromatic number of the pairwise conflict graph; see section 8.

From Groups to Configurations and the GSP

Given a grouping $G_1, \dots, G_K$, each group G_k must be served by a *configuration* C_k (a magazine state with $|C_k| = b$) that covers it.

Definition 1.4 (Configuration cluster and slack). For a group G with required set $T^{(G)} = \bigcup_{j \in G} T_j$,

$$\mathcal{C}(G) = \{C \subseteq T : |C| = b, T^{(G)} \subseteq C\}, \quad s(G) = b - |T^{(G)}| \geq 0.$$

The $s(G)$ free slots (“slack”) may be filled with tools of *adjacent* groups to lower the transition cost.

Definition 1.5 (Group Sequencing Problem and the heuristic cost H). The *GSP* chooses, for a given grouping, a permutation π of the groups and a configuration $C_k \in \mathcal{C}(G_k)$ for each k , minimising

$$H(\pi, \{C_k\}) = \sum_{i=1}^{K-1} |C_{\pi(i+1)} \setminus C_{\pi(i)}| = \sum_{i=1}^{K-1} (b - |C_{\pi(i)} \cap C_{\pi(i+1)}|).$$

This is a path **Generalised TSP** over the clusters $\{\mathcal{C}(G_k)\}$. The JGP + GSP *heuristic* (1) solves the JGP to get a K^* -grouping, then (2) solves the GSP exactly; its cost is

$$H = \min_{\text{JGP-optimal groupings}} \min_{\pi, \{C_k\}} H(\pi, \{C_k\}),$$

the minimum over all K^* -group JGP solutions of the GTSP optimum. Any such schedule is a feasible SSP solution, so $H \geq Z_{\text{SSP}}^*$.

Lemma 1.6 (Per-transition lower bound). *For an ordered adjacent pair G_k, G_{k+1} , every choice of configurations costs at least*

$$w^*(G_k, G_{k+1}) = \max\left(0, |T^{(G_k)} \cup T^{(G_{k+1})}| - b\right).$$

For zero-slack groups ($s(G_k) = s(G_{k+1}) = 0$) the configurations are forced ($C_k = T^{(G_k)}$) and the transition costs exactly $b - |T^{(G_k)} \cap T^{(G_{k+1})}|$.

Proof. $C_{k+1} \supseteq T^{(G_{k+1})}$ has $s(G_{k+1})$ free slots; of the $|T^{(G_k)} \setminus T^{(G_{k+1})}|$ tools that C_k holds but G_{k+1} does not require, at most $s(G_{k+1})$ can be retained, so at least $\max(0, |T^{(G_k)} \setminus T^{(G_{k+1})}| - s(G_{k+1})) = \max(0, |T^{(G_k)} \cup T^{(G_{k+1})}| - b)$ must leave. With zero slack no tool can be retained beyond the required set, giving the stated exact cost. $\square$

Greedy “lookahead” is not globally optimal for the GSP. Filling each group’s slack with the next group’s tools (lookahead) attains the per-transition bound of lemma 1.6 for one isolated transition, but the padding of C_k influences *both* the $(k-1, k)$ and $(k, k+1)$ transitions, so summing per-transition minima can under-count. The true heuristic cost H is the GTSP optimum of definition 1.5, which we compute exactly. For *zero-slack* instances there is no padding freedom and the GSP reduces to a TSP over forced configurations; all worst-case instances in this note are zero-slack, where the per-transition formula is exact.

Lower and Upper Bounds Relating Z_{SSP}^* , K^* and the Tool Count

Two elementary but distinct lower bounds control Z_{SSP}^* . Both are used repeatedly.

Theorem 2.1 (Grouping lower bound). $Z_{\text{SSP}}^* \geq K^* - 1$.

Proof. An optimal SSP solution visits a sequence of magazine states; let $W_1, \dots, W_m$ be the distinct states in order of first appearance. Each job is served by some W_r , so $\{W_1, \dots, W_m\}$ covers J by feasible configurations and $m \geq K^*$. Consecutive distinct states differ in ≥ 1 tool, so the number of insertions is $\geq m - 1 \geq K^* - 1$. $\square$

Theorem 2.2 (Tool-counting lower bound). *Let $U = \bigcup_{j \in J} T_j$. Then $Z_{\text{SSP}}^* \geq |U| - b$.*

Proof. Every tool of U is required by some job, hence is loaded into the magazine at least once. At most b tools occupy the initial free load; every other loading is a counted insertion. Therefore the number of insertions is at least $|U| - b$. $\square$

Remark 2.3 (Relation to the SSPMF bound). $|U| - b$ is exactly the $M - C$ linear-programming lower bound of the multicommodity-flow formulation (SSPMF) of da Silva et al. [da Silva and Yanasse(2024)] ($M = |U|$ tools, $C = b$ capacity — the identification $M = |U|$ assumes every tool is required by

some job, w.l.o.g. after deleting unused tools); theorem 2.2 gives the elementary combinatorial proof. Combining the two bounds,

$$Z_{\text{SSP}}^* \geq \max(K^* - 1, |U| - b).$$

Theorem 2.4 (Grouping upper bound). $Z_{\text{SSP}}^* \leq b(K^* - 1)$.

Proof. Process a K^* -group JGP solution batch by batch under one fixed covering configuration per batch. The initial load is free; each of the $K^* - 1$ inter-batch transitions inserts at most b tools. $\square$

Which bound is tight? Neither lower bound is tight in general, and their maximum need not equal Z_{SSP}^* . On the ring family of section 3 and on the entire exhaustively enumerated $b = 3$ edge family (jobs = graph edges, $|T_j| = 2$; 10691 instances) we verified $Z_{\text{SSP}}^* = \max(K^* - 1, |U| - b)$ with no exception. But this equality is *family-specific*: among $b = 3$ instances with $|T_j| \leq 3$ on ≤ 5 tools, 2410 of 21569 have Z_{SSP}^* strictly above *both* bounds, and the $b = 4$ sliding window $T_{j_i} = \{i, i+1, i+2\} \bmod 8$ has $Z_{\text{SSP}}^* = 5 > \max(3, 4)$. Only the inequality of remark 2.3 is universal.

A Certified Counterexample: the 6-Job Ring

Example 3.1 (6-job ring). Let $T = \{t_1, \dots, t_6\}$, $b = 3$, and $T_{j_i} = \{t_i, t_{i+1 \bmod 6}\}$ for $i = 1, \dots, 6$: consecutive jobs share one tool.

JGP optimum ($K^* = 3$). The minimum number of feasible groups is 3 (two optimal groupings exist). One is $G_1 = \{j_1, j_2\}$, $G_2 = \{j_3, j_4\}$, $G_3 = \{j_5, j_6\}$ with required sets $T^{(1)} = \{t_1, t_2, t_3\}$, $T^{(2)} = \{t_3, t_4, t_5\}$, $T^{(3)} = \{t_1, t_5, t_6\}$, each of size 3 (zero slack), so configurations are forced.

JGP + GSP cost ($H = 4$). With forced configurations the GSP is a TSP on three nodes; every pair of the three sets shares exactly one tool, so each transition costs $b - 1 = 2$ and any order pays $2 \times 2 = 4$. Thus $H = 4$.

SSP optimum ($Z_{\text{SSP}}^* = 3$). The sequence $j_1 \rightarrow \dots \rightarrow j_6$ under KTNS:

Step	Job	Magazine	Switches
1	j_1	$\{t_1, t_2, t_3\}$	(free initial load)
2	j_2	$\{t_1, t_2, t_3\}$	0
3	j_3	$\{t_1, t_3, t_4\}$	1 ($-t_2, +t_4$)
4	j_4	$\{t_1, t_4, t_5\}$	1 ($-t_3, +t_5$)
5	j_5	$\{t_1, t_5, t_6\}$	1 ($-t_4, +t_6$)
6	j_6	$\{t_1, t_5, t_6\}$	0

KTNS retains t_1 as a *bridge* across jobs j_2 – j_5 , so $Z_{\text{SSP}}^* = 3$, matching both lower bounds: $|U| - b = 6 - 3 = 3$ (and $K^* - 1 = 2$).

Gap. $H - Z_{\text{SSP}}^* = 4 - 3 = 1$; ratio $4/3$. Verified exactly by state-DP and by PORTA enumeration.

The Gap is Unbounded

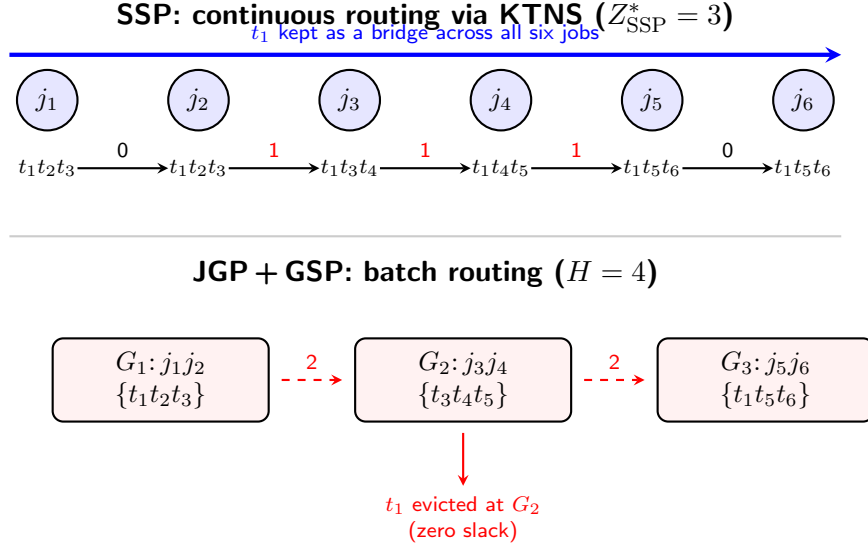


Figure 1: The 6-ring. KTNS keeps t_1 resident through all jobs (3 switches). The 3-group JGP saturates each batch with its required tools, forcing t_1 out at G_2 and back in at G_3 (4 switches). Gap = 1.

Theorem 4.1 (Gap grows with K^*). *For every integer $g \geq 1$ there is a uniform-SSP instance with $K^* = 3g$ on which the JGP + GSP heuristic costs exactly g more switches than the optimum: $Z_{\text{SSP}}^* = 6g - 3$, $H = 7g - 3$, gap = g .*

Proof. Take g tool-disjoint copies of the 6-ring (example 3.1); copy ℓ uses tools $\{t_{6\ell-5}, \dots, t_{6\ell}\}$, so $|U| = 6g$.

Optimum. By theorem 2.2, $Z_{\text{SSP}}^* \geq |U| - b = 6g - 3$. Processing the copies contiguously achieves this value (within a copy, 3 switches as in example 3.1; between two copies the magazine swaps completely, costing $b = 3$, of which the first such load per copy realises that copy's tool count); the explicit schedule attains $6g - 3$, so $Z_{\text{SSP}}^* = 6g - 3$.

Heuristic. The clusters of different copies are tool-disjoint, so every inter-copy transition costs $b = 3$ and the GTSP decomposes: $H = \sum_{\ell} H^{(\ell)} + b(g - 1) = 4g + 3(g - 1) = 7g - 3$.

Hence gap = $H - Z_{\text{SSP}}^* = (7g - 3) - (6g - 3) = g$. $\square$

Remark 4.2 (The gap scales with K^* , not b). Here gap = $g = K^*/3$, consistent with the worst-case envelope gap $\leq H \leq b(K^* - 1)$ from theorem 2.4. The relevant complexity parameter for the additive gap is the number of groups K^* , not the capacity b . The ratio, by contrast, *shrinks* as g grows: $(7g - 3)/(6g - 3)$ decreases from $4/3$ at $g = 1$ to $7/6$ as $g \rightarrow \infty$ (section 5). All values verified for $g = 1, \dots, 4$.

Remark 4.3 (Two informal “sources” of gap). It is tempting to split the gap into a *padding* part (even the best configurations for a fixed K^* -grouping cost more than Z_{SSP}^*) and a *grouping* part (the SSP optimum need not respect any K^* -grouping). This is a useful intuition but not a theorem: the two effects are not independent and there is no proved decomposition. In the 6-ring, exhaustive enumeration of all 18 feasible groupings shows the minimum GTSP cost over K^* -groupings is 4 while over *all* groupings it is 3; thus for this instance the entire heuristic gap is removed by changing the grouping, not the padding — see theorem 6.6.

A Computable Gap Bound, the Bridge Mechanism, and the Ratio

A directly computable gap bound

Proposition 5.1 (Gap bound). *For every instance,*

$$0 \leq \text{gap}(I) = H - Z_{\text{SSP}}^* \leq H - \max(K^* - 1, |U| - b).$$

Proof. $H \geq Z_{\text{SSP}}^*$ because the heuristic schedule is SSP-feasible; the upper bound substitutes the proved lower bound $Z_{\text{SSP}}^* \geq \max(K^* - 1, |U| - b)$ (remark 2.3). $\square$

For the 6-ring this gives $\text{gap} \leq 4 - \max(2, 3) = 1$, which is tight. The bound is computable without solving the SSP, since H , K^* and $|U|$ are all available from the heuristic.

The bridge / re-entry mechanism

The gap is caused by tools the SSP keeps resident across a group boundary but the heuristic must evict and reload.

Definition 5.2 (Bridge tool). Fix a K^* -grouping in GSP-optimal order. A tool t is a *bridge* if it is required in some group G_k and again in some later $G_{k'}$ ($k' > k$) but not required by an intermediate group whose forced configuration must therefore evict it. KTNS can keep t resident through the intermediate jobs using slack, saving one insertion that the heuristic pays.

In the 6-ring, t_1 is required by $j_1 \in G_1$ and $j_6 \in G_3$ but not by $G_2 = \{j_3, j_4\}$ (whose forced configuration is $\{t_3, t_4, t_5\}$). KTNS keeps t_1 through $j_2, \dots, j_5$; the heuristic evicts it at the $G_1 \rightarrow G_2$ boundary and reloads it at $G_2 \rightarrow G_3$, which is the one extra switch.

Retraction. An earlier draft claimed an intermediate inequality $\sum_k R_k \leq H/4$ (with R_k the number of re-entering tools at boundary k) and chained it to $H/Z_{\text{SSP}}^* \leq 4/3$. The intermediate inequality is **false**: the zero-slack triple $T^{(1)} = \{a, b, c\}$, $T^{(2)} = \{a, b, d\}$, $T^{(3)} = \{c, d, e\}$ ($b = 3$) in GSP order has $H = 3$ and $\sum_k R_k = 1 > H/4 = 0.75$ (verified). Hence *there is no proof of the 4/3 ratio along this route*; the ratio bound below is stated as a conjecture.

The approximation ratio for $b = 3$

Proposition 5.3 (Ratio lower bound). $\sup_{I: b=3} \frac{H(I)}{Z_{\text{SSP}}^*(I)} \geq \frac{4}{3}$, *witnessed by the 6-ring* ($H = 4$, $Z_{\text{SSP}}^* = 3$).

Conjecture 5.4 (Tight ratio for $b = 3$). *For all uniform-SSP instances with $b = 3$, $\frac{H}{Z_{\text{SSP}}^*} \leq \frac{4}{3}$, with equality only at the 6-ring (up to relabelling).*

Status of conjecture 5.4: open, strong computational support.

- Exhaustive search over the $b = 3$ edge family (jobs = edges of a graph on ≤ 6 tool-

vertices, ≤ 6 edges; 10691 instances): the maximum ratio observed is exactly $4/3$ and is *never* exceeded; every positive-gap instance there has $K^* = 3$, $|U| = 6$, $Z_{\text{SSP}}^* = 3$, $H = 4$.

- The unbounded-gap family (theorem 4.1) has ratios in $[7/6, 4/3]$, supremum $4/3$ at $g = 1$.
- Rings of length $k = 6, 7, 8$ give ratios $4/3, 5/4, 6/5$; shorter rings have gap 0.

No $b = 3$ instance with ratio $> 4/3$ is known. A proof (or refutation) is the main open problem of this note; section 7 gives a general- b programme (unconditional bound theorem 7.3, conjecture 7.6, and a conditional route to $4/3$, proposition 7.8). For $b \geq 4$ the extremal ratio is *not* bounded by $4/3$: a broadened search reaches (exactly) $4/3$ at $b = 4$ and $3/2$ at $b = 5$, so $\sup_I H/Z_{\text{SSP}}^*$ appears to grow with b — the $4/3$ bound is a $b = 3$ phenomenon.

When is the Gap Zero?

Since $H \geq Z_{\text{SSP}}^* \geq \max(K^* - 1, |U| - b)$, the heuristic is optimal whenever it meets either lower bound. The next theorem, however, is the real organising principle: it shows the gap is *exactly* the cost of the heuristic's insistence on the minimum number of groups.

Grouping exactness: the gap is the price of minimum K^*

For a grouping $\mathcal{G}$ (any partition of J into feasible groups, of any cardinality), write $\text{GTSP}(\mathcal{G})$ for its path-GTSP optimum (definition 1.5).

Theorem 6.1 (Grouping exactness). *For every uniform-SSP instance,*

$$Z_{\text{SSP}}^* = \min_{\mathcal{G} \text{ feasible}} \text{GTSP}(\mathcal{G}),$$

the minimum being over all feasible groupings. Consequently the JGP + GSP gap is

$$H - Z_{\text{SSP}}^* = \underbrace{\min_{|\mathcal{G}|=K^*} \text{GTSP}(\mathcal{G})}_{=H} - \underbrace{\min_{\mathcal{G}} \text{GTSP}(\mathcal{G})}_{=Z_{\text{SSP}}^*},$$

i.e. precisely the penalty for restricting the heuristic to JGP-optimal (minimum-cardinality) groupings. With unrestricted grouping choice the framework is exact.

Proof. ($\leq$) Any feasible grouping together with a GSP ordering and one configuration per group is a feasible SSP schedule of cost $\text{GTSP}(\mathcal{G})$, so $Z_{\text{SSP}}^* \leq \min_{\mathcal{G}} \text{GTSP}(\mathcal{G})$. ($\geq$) Let σ be an optimal SSP schedule with full-magazine trajectory $M_1, \dots, M_n$ (one state per job; full w.l.o.g.). Bring it into *normal form*: collapse consecutive equal states; if a configuration recurs at a later, non-consecutive visit, reassign the jobs served there to its first visit (the configuration is identical) and delete the later visit, joining its neighbours $X \rightarrow C \rightarrow Y$ into $X \rightarrow Y$ — by the triangle inequality $|Y \setminus X| \leq |C \setminus X| + |Y \setminus C|$ the cost does not increase; delete any state serving no job by the same shortcut. The result is a sequence of pairwise-distinct configurations $C_{(1)}, \dots, C_{(m)}$, each serving at least one job; it is itself a feasible schedule of cost $\leq Z_{\text{SSP}}^*$, hence exactly Z_{SSP}^* by optimality. Assign each job to the configuration serving it: the jobs of $C_{(i)}$ have tool union contained in $C_{(i)}$, hence form a feasible group, and these groups partition J . The configurations

$C_{(1)}, \dots, C_{(m)}$ in order constitute a GTSP path over this grouping of cost exactly Z_{SSP}^* , so $\min_{\mathcal{G}} \text{GTSP}(\mathcal{G}) \leq Z_{\text{SSP}}^*$. Equality follows. (Confirmed on 1306 instances with no exception; `plans-genai/_verification/study_op.py`.) $\square$

Remark 6.2 (Interpretation). theorem 6.1 relocates the entire difficulty of the SSP into a single combinatorial choice: *which* grouping to use. The JGP solves for the *fewest* groups; the SSP wants the *cheapest-to-sequence* groups, which may be more numerous. The bridge tools of definition 5.2 are exactly the tools that a finer grouping can keep resident. This is the precise sense in which “the gap is the price of minimum K^* ”, and it motivates optimising over groupings directly (Part VI).

The heuristic meets a lower bound

Theorem 6.3 (Sufficient condition A — grouping bound met). *If some JGP-optimal grouping admits a group order whose GTSP cost equals $K^* - 1$, then $\text{gap} = 0$ and $Z_{\text{SSP}}^* = H = K^* - 1$.*

Proof. $H \leq K^* - 1 \leq Z_{\text{SSP}}^* \leq H$ by theorem 2.1 and $H \geq Z_{\text{SSP}}^*$; all are equal. $\square$

Theorem 6.4 (Sufficient condition B — tool bound met). *If some JGP-optimal grouping has GTSP cost equal to $|U| - b$, then $\text{gap} = 0$ and $Z_{\text{SSP}}^* = H = |U| - b$.*

Proof. $H \leq |U| - b \leq Z_{\text{SSP}}^* \leq H$ by theorem 2.2 and $H \geq Z_{\text{SSP}}^*$. $\square$

Remark 6.5 (Verified picture for $b = 3$). Across the 10691-instance edge family, every zero-gap instance meets one of the two bounds (the heuristic GTSP cost equals $\max(K^* - 1, |U| - b)$), and every positive-gap instance has $K^* = 3$. So $K^* \leq 2$ already forces $\text{gap} = 0$ in this family (now a theorem for all instances and all b : corollary 7.4). A full structural characterisation of the zero-gap set (a necessary-and-sufficient condition) is open.

Sub-optimal groupings can beat the JGP-optimal heuristic

The heuristic insists on a K^* -grouping. By theorem 6.1 some grouping always attains Z_{SSP}^* , and when no K^* -grouping does, a larger one must — relaxing $|\mathcal{G}| = K^*$ strictly helps. The following is the smallest explicit witness.

Theorem 6.6 (A sub-optimal grouping closes the ring gap). *For the 6-ring, the $K = 4$ grouping $\{j_1, j_2\}, \{j_3, j_4\}, \{j_5\}, \{j_6\}$ admits configurations and an order with total GTSP cost $3 = Z_{\text{SSP}}^*$. Hence a grouping with $K > K^*$ achieves cost strictly below the JGP-optimal heuristic value $H = 4$.*

Proof. The singletons $\{j_5\}$ and $\{j_6\}$ have slack 1, so their configurations can retain a non-required “bridge” tool. Order the four groups and pad as

$$\underbrace{\{t_1, t_2, t_3\}}_{\{j_1, j_2\}} \rightarrow \underbrace{\{t_1, t_3, t_6\}}_{\{j_6\}} \rightarrow \underbrace{\{t_3, t_5, t_6\}}_{\{j_5\}} \rightarrow \underbrace{\{t_3, t_4, t_5\}}_{\{j_3, j_4\}}.$$

Each configuration covers its group ($\{j_6\}$ needs $\{t_1, t_6\}$, $\{j_5\}$ needs $\{t_5, t_6\}$), and consecutive configurations share two tools, so every transition costs 1: total $3 = Z_{\text{SSP}}^*$ (the tool t_3 now plays the bridge role across $\{j_6\}, \{j_5\}$). Exhaustive evaluation of all 18 feasible groupings (script `run_study.py`) confirms the minimum GTSP cost over $K = 4$ groupings is 3, versus 4 over all $K^* = 3$ groupings. $\square$

Remark 6.7 (Consequence). This answers, affirmatively, the question of whether sub-optimal JGP solutions can lower the SSP cost of the decomposition: *yes*. It also refutes the plausible-looking claim that “extra groups can only add transitions” for zero-slack instances — splitting a zero-slack group creates slack in the parts, which can enable bridging the heuristic otherwise loses. The right object to optimise is therefore not Z_{JGP}^* but the joint (grouping, GSP) cost; this is the subject of Part VI (`06_grouping_selection.tex`).

General- b Bounds on the Ratio

Throughout this section $m = |U|$, $q = m - b$, and $K^* \geq 2$ (the case $K^* = 1$ is trivial, $H = Z_{\text{SSP}}^* = 0$). Recall $Z_{\text{SSP}}^* \geq \max(K^* - 1, q)$ (remark 2.3).

The cost identity and re-insertions

Lemma 7.1 (Cost identity). *W.l.o.g. every magazine state of a full-magazine schedule (and every GSP configuration) is a subset of U (pad within U ; possible since $m \geq b$ when $K^* \geq 2$). Then the cost of any such schedule equals*

$$\text{cost} = (m - b) + R,$$

where $R \geq 0$ counts the re-insertions: insertions of tools previously present and evicted (including evicted tools of the free initial load).

Proof. The initial load holds b tools of U for free. Each of the remaining $m - b$ tools of U is required by some job, hence inserted at least once. Splitting the total insertions into first-time insertions (exactly $m - b$) and all others (by definition R) gives the identity. $\square$

Consequently $Z_{\text{SSP}}^* = q + R_{\text{opt}}$ and $H = q + R_H$, so $\text{gap} = R_H - R_{\text{opt}}$: the gap is exactly the excess re-insertions forced by the K^* -grouping. This makes the bridge mechanism of definition 5.2 quantitative.

Unconditional bounds

Lemma 7.2 (Transition cap). *In any GSP path with configurations $\subseteq U$, every transition costs $|C_{k+1} \setminus C_k| \leq \min(b, q)$. Hence $H \leq (K^* - 1) \cdot \min(b, q)$.*

Proof. $|C_{k+1} \setminus C_k| \leq |C_{k+1}| = b$ and $|C_{k+1} \setminus C_k| \leq |U \setminus C_k| = m - b = q$ since $C_{k+1} \subseteq U$ and $|C_k| = b$. A covering configuration within U exists for every group, so the heuristic may be restricted to such configurations w.l.o.g. $\square$

Theorem 7.3 (Unconditional ratio bound). *For every uniform-SSP instance with $K^* \geq 2$,*

$$\frac{H}{Z_{\text{SSP}}^*} \leq \frac{(K^* - 1) \min(b, q)}{\max(K^* - 1, q)} \leq \min(b, K^* - 1, q).$$

Proof. $Z_{\text{SSP}}^* \geq \max(K^* - 1, q) \geq 1$, so divide lemma 7.2 by it. For the second inequality: $(K^* - 1)q / \max(K^* - 1, q) = \min(K^* - 1, q)$ exactly, and $\min(b, q) \leq q$ gives the $\min(K^* - 1, q)$ part, while $(K^* - 1) / \max(K^* - 1, q) \leq 1$ and $\min(b, q) \leq b$ give the b part. $\square$

Corollary 7.4 (Zero-gap classes, all b). *(i) If $K^* = 2$ then $\text{gap} = 0$ (the ratio bound is $\leq K^* - 1 = 1$); in fact $H = Z_{\text{SSP}}^* = m - b$, attained by padding C_1 with tools of $T^{(2)}$ and C_2 with tools of C_1 . (ii) If $m \leq b + 1$ then $\text{gap} = 0$. (iii) If $K^* = 3$ then $H/Z_{\text{SSP}}^* \leq 2$.*

Proof. All three are instances of theorem 7.3 ($K^* - 1 = 1$; $q \leq 1$; $K^* - 1 = 2$). For the explicit construction in (i): $C_1 = T^{(1)} \cup P$ with $P \subseteq T^{(2)} \setminus T^{(1)}$, $|P| = b - |T^{(1)}|$ (possible since $m \geq b$), and $C_2 = T^{(2)} \cup (\text{pad from } C_1)$; then $|C_2 \setminus C_1| = |T^{(2)} \setminus C_1| = (m - |T^{(1)}|) - (b - |T^{(1)}|) = m - b \leq Z_{\text{SSP}}^* \leq H$. $\square$

Remark 7.5. corollary 7.4(i) upgrades the empirical observation of section 6 (“ $K^* \leq 2$ forces $\text{gap} = 0$ in the $b = 3$ edge family”) to a theorem for *all* instances and *all* b . Worst cases therefore require $K^* \geq 3$, consistent with every positive-gap instance found so far having $K^* \in \{3, 4\}$.

The excess re-insertion conjecture

All known extremal instances satisfy a much stronger bound than theorem 7.3, with *equality*:

Conjecture 7.6 (Excess re-insertions; generalises the $K^*=3, b=3$ conjecture). *For every uniform-SSP instance with $K^* \geq 2$,*

$$\text{gap} = H - Z_{\text{SSP}}^* \leq K^* - 2.$$

Evidence for conjecture 7.6. $K^* = 2$ is *proved* (corollary 7.4). Every known extremal witness attains the bound with equality: the 6-ring ($K^*=3, \text{gap}=1$), the $b=4$ witness ($K^*=3, \text{gap}=1$), the $b=5$ witness ($K^*=4, \text{gap}=2$); the g -copy family has $\text{gap} = g \leq 3g - 2$. No violation in ~ 1380 instances ($b \in \{2, 3, 4\}$, mixed job sizes; `_verification/verify_ratio_section.py`), nor anywhere in the earlier exhaustive searches. The first open case is $K^* = 3$ for general b : $\text{gap} \leq 1$.

Proposition 7.7 (Consequences of conjecture 7.6). *If conjecture 7.6 holds, then for every instance and every b :*

$$\frac{H}{Z_{\text{SSP}}^*} \leq 1 + \frac{K^* - 2}{\max(K^* - 1, q)} \leq 2 - \frac{1}{K^* - 1} < 2,$$

i.e. the JGP + GSP heuristic is a $(2 - \varepsilon)$ -approximation universally in b . Moreover all

known extremal witnesses have $q = K^*$ and $\text{gap} = K^* - 2$, where the first bound evaluates to $2 - 2/K^*$ — reproducing exactly the observed extremal ratios $4/3$ ($K^*=3$) and $3/2$ ($K^*=4$).

Proof. Divide $H \leq Z_{\text{SSP}}^* + K^* - 2$ by $Z_{\text{SSP}}^* \geq \max(K^* - 1, q)$; the second inequality takes $q \geq 1$. At $q = K^*$: $1 + (K^* - 2)/K^* = 2 - 2/K^*$. $\square$

Proposition 7.8 (A route to the $4/3$ conjecture). Assume (a) conjecture 7.6, and (b) every positive-gap $b = 3$ instance has $K^* = 3$ and $Z_{\text{SSP}}^* \geq 3$ (verified exhaustively on the 10691-instance edge family, where positive gap forces $K^* = 3$, $|U| = 6$, $Z_{\text{SSP}}^* = 3$; open in general). Then conjecture 5.4 holds: $H/Z_{\text{SSP}}^* \leq 4/3$ for $b = 3$.

Proof. Zero-gap instances have ratio 1. For a positive-gap instance, (a) and (b) give $\text{gap} \leq K^* - 2 = 1$ and $H/Z_{\text{SSP}}^* = 1 + \text{gap}/Z_{\text{SSP}}^* \leq 1 + 1/3 = 4/3$. $\square$

Open problems raised by this section.

- *Prove conjecture 7.6*, starting with $K^* = 3$ (claim: $\text{gap} \leq 1$ for all b). By lemma 7.1 this says: some JGP-optimal grouping and order forces at most $K^* - 2$ more re-insertions than the SSP optimum.
- *Achievability of ratio 2*. proposition 7.7 caps the ratio at $2 - 1/(K^* - 1)$; approaching 2 needs $K^* \rightarrow \infty$ with $q \approx K^* - 1$ and near-maximal gap *simultaneously*. The g -copy construction fails: its tools decouple, Z_{SSP}^* grows linearly per copy, and the ratio *decreases* to $7/6$. A family with a *shared* bridge structure across unboundedly many groups (or a proof that none exists, yielding a bound strictly below 2) would settle $\sup_b \sup_I H/Z_{\text{SSP}}^*$.
- *The per- b extremal curve*. Known suprema: $4/3$ ($b=3$), $\geq 4/3$ ($b=4$), $\geq 3/2$ ($b=5$), all of the form $2 - 2/K^*$ at witnesses with $q = K^*$. Determine $\sup_I H/Z_{\text{SSP}}^*$ as a function of b ; in particular whether $q = K^* - 1$ extremal instances (which would give the larger value $2 - 1/(K^* - 1)$) exist for any b .

Conflict-Graph Structure: What It Does and Does Not Predict

Define the *pairwise conflict graph* G_{conf} on the jobs, with an edge (j, j') whenever $|T_j \cup T_{j'}| > b$. Every feasible grouping is a proper colouring of G_{conf} (a group contains no conflicting pair), so

$$\chi(G_{\text{conf}}) \leq K^*.$$

Remark 8.1 (K^* is not the chromatic number). The inequality can be strict. For the star $\{t_1, t_2\}, \{t_1, t_3\}, \{t_1, t_4\}$ ($b = 3$) we have $\chi(G_{\text{conf}}) = 1$ (no conflicting pair) but $K^* = 2$ (the joint union has size 4). Equality $\chi = K^*$ holds when the tool sets enjoy a Helly-type property under which pairwise compatibility implies joint feasibility — e.g. the ring instances below. The identification “ $K^* = \chi(G_{\text{conf}})$ ” is therefore valid only under such a hypothesis, not in general.

Remark 8.2 (Conflict-graph structure of the rings, corrected). For the 6-ring, two jobs conflict iff they are tool-disjoint, i.e. non-adjacent on the ring. The resulting graph is 3-regular with 9 edges: it is the *triangular prism* $\overline{C_6}$ (the complement of the 6-cycle), **not** $2K_3$. (Earlier notes stated $2K_3$; that is incorrect — e.g. $j_1 = \{t_1, t_2\}$ and $j_4 = \{t_4, t_5\}$ are disjoint and hence *do* conflict.) The prism is perfect, being the complement of an even cycle. For the 5-ring the conflict graph is the imperfect odd cycle C_5 ($\chi = 3$, $\chi_f = 5/2$).

Remark 8.3 (Neither perfectness nor the JGP LP gap predicts the SSP gap). The 6-ring has a *perfect* conflict graph (the prism) yet $\text{gap} = 1$; the 5-ring has an *imperfect* conflict graph (C_5) yet $\text{gap} = 0$ (here $K^* = 3$, $Z_{\text{SSP}}^* = 2$, $H = 2$, both lower bounds met). Likewise the JGP LP relaxation value $\chi_f(G_{\text{conf}})$ does not track the SSP gap: for the 5-ring $\chi_f = 5/2 < 3 = K^*$ (a fractional JGP gap) but the SSP gap is 0, whereas for the 6-ring $\chi_f = 3 = K^*$ (no fractional JGP gap) but the SSP gap is 1. Thus the SSP gap is governed by tool-overlap structure at group boundaries (the bridge mechanism of definition 5.2), not by global chromatic or perfectness properties. A positive structural characterisation remains open. References on perfection and clutters: [Chudnovsky et al.(2006)Chudnovsky, Robertson, Seymour, and Thomas, Cornuéjols(2001)].

Connections to Solution Methods

The quantities here interface with the exact methods developed in the other Parts but are logically separate from them:

- the GSP is a path-GTSP, solvable by GTSP/TSP machinery (Noon–Bean, LKH, branch-and-cut) as surveyed in [Pop et al.(2024)Pop, Cosma, Sabo, and Pop Sitar]; this is the engine of phase 2;
- the non-compact configuration ILP and its column generation — including why a Dantzig–Wolfe master over job sequences degenerates, and the set-covering CG for the JGP — belong to Part II (`02_noncompact_formulation.tex`);
- the MTZ alternative to subtour constraints is Part IV (`04_mtz_formulation.tex`);
- the branch-and-Benders exact solver (master TSP + KTNS subproblem) is Part VIII (`08_branch_and_benders.tex`), which reaches the same Z_{SSP}^* from the sequence side.

The bounds of section 2 (especially $Z_{\text{SSP}}^* \geq \max(K^* - 1, |U| - b)$) are the natural primal bounds to seed any of these solvers.

References

- [Catanzaro et al.(2015)Catanzaro, Gouveia, and Labbé] D. Catanzaro, L. Gouveia, and M. Labbé. Improved integer linear programming formulations for the job sequencing and tool switching problem. *European Journal of Operational Research*, 244(3):766–777, 2015.
- [Chudnovsky et al.(2006)Chudnovsky, Robertson, Seymour, and Thomas] M. Chudnovsky, N. Robertson, P. Seymour, and R. Thomas. The strong perfect graph theorem. *Annals of Mathematics*, 164(1):51–229, 2006.
- [Cornuéjols(2001)] G. Cornuéjols. *Combinatorial Optimization: Packing and Covering*. Number 74 in CBMS-NSF Regional Conference Series in Applied Mathematics. SIAM, 2001.
- [Crama and Oerlemans(1994)] Y. Crama and A. G. Oerlemans. A column generation approach to job grouping for flexible manufacturing systems. *European Journal of Operational Research*, 78(1):58–80, 1994.
- [da Silva and Yanasse(2024)] A. A. C. da Silva and H. H. Yanasse. A multicommodity flow model and exact methods for the job sequencing and tool switching problem. *Preprint (compiled 2024)*; see also *Computers & Operations Research*, 2024. SSPMF multicommodity-flow formulation; LP lower bound $M - C$.
- [Otiai(2024)] F. S. Otiai. Efficient tool management in flexible manufacturing: A branch and price approach to the job grouping problem. Master’s thesis, ISIMA, Clermont Auvergne INP, 2024.

- [Pop et al.(2024)Pop, Cosma, Sabo, and Pop Sitar] P. C. Pop, O. Cosma, C. Sabo, and C. Pop Sitar. The generalized traveling salesman problem: A comprehensive review. *European Journal of Operational Research*, 314(3):819–835, 2024.
- [Tang and Denardo(1988)] C. S. Tang and E. V. Denardo. Models arising from a flexible manufacturing machine, part i: Minimization of the number of tool switches. *Operations Research*, 36(5):767–777, 1988.